

11(a)	packet/quantum of <u>energy</u>	M1
	of electromagnetic radiation	A1
11(b)(i)	$E = hc / \lambda$	C1
	$1.18 \times 1.60 \times 10^{-13} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	A1
	$\lambda = 1.05 \times 10^{-12} \text{ m}$	
11(b)(ii)	$\lambda = h / p$ or $E = pc$	C1
	$p = (6.63 \times 10^{-34}) / (1.05 \times 10^{-12})$ or $p = (1.18 \times 1.60 \times 10^{-13}) / (3.00 \times 10^8)$	B1
	leading to $p = 6.3 \times 10^{-22} \text{ N s}$	
11(c)	$6.3 \times 10^{-22} = 60 \times 1.66 \times 10^{-27} \times v$	C1
	$v = 6.3 \times 10^3 \text{ m s}^{-1}$	A1